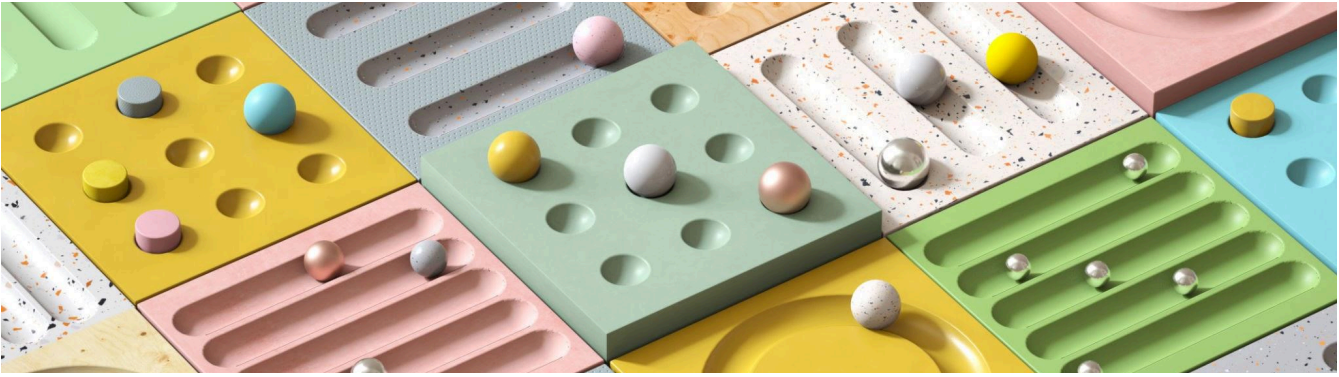
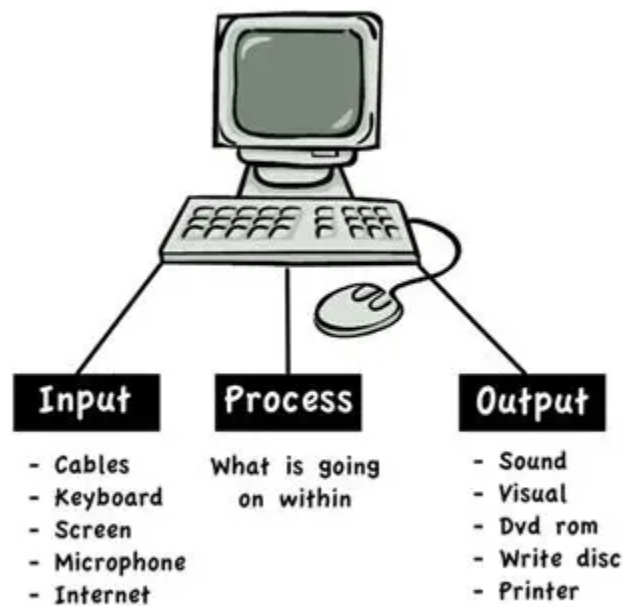


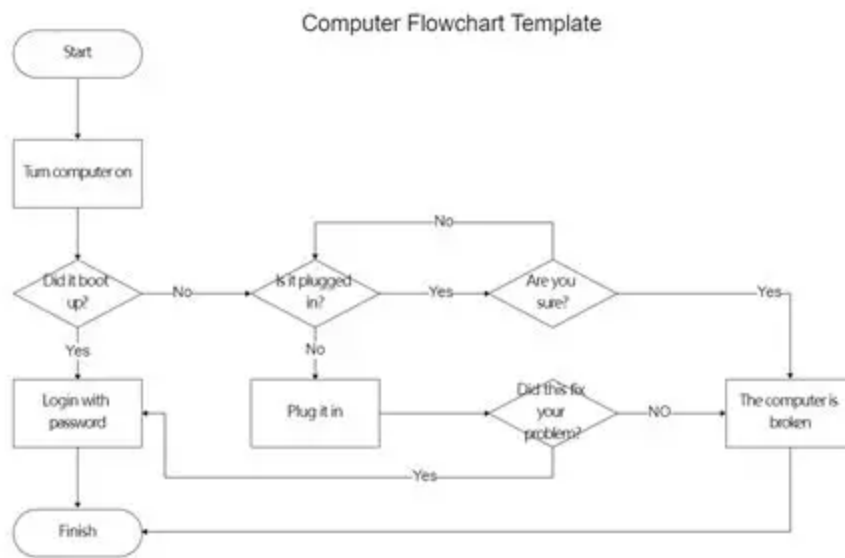


Chapter 1 – Introduction to Computers

1.1 Definition of a Computer

A computer follows the cycle: Input → Process → Output → Storage.

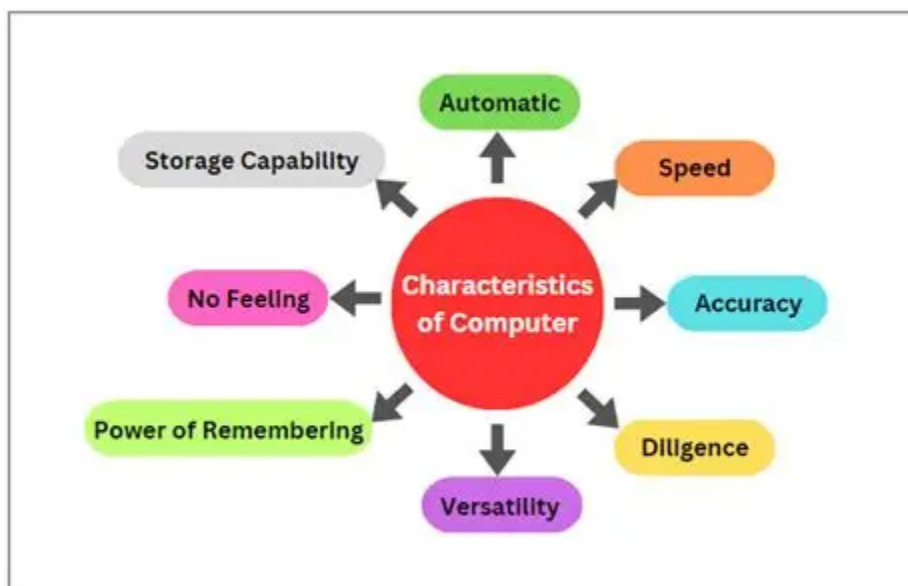




💡 *Example:* Typing a document → CPU processes → Monitor shows → File saved on disk.

1.2 Characteristics of Computers

- Speed ⚡ – Millions of instructions per second.
- Accuracy ✔️ – Executes tasks without mistakes.
- Automation 🤖 – Works once programmed.
- Storage 📦 – Gigabytes to terabytes.
- Versatility 🔄 – Banking, education, entertainment.
- Connectivity 🌐 – Internet access worldwide.



1.3 Types of Computers

- Supercomputers 🖨️ – Weather forecasting.
- Mainframes 🏢 – Banking, railways.
- Minicomputers 🏭 – Industry use.
- Microcomputers 💻 – PCs, laptops.
- Mobile Devices 📱 – Smartphones, tablets.



Microcomputer VS Supercomputer

Comparison Chart

Microcomputer	Supercomputer
Microcomputers are the general purpose computers mostly used for day to day work.	Supercomputer refers to the class of extremely powerful computers that are able to perform at or near the highest operational rate.
A microcomputer is a computer system designed to meet the information processing needs of a single person.	Supercomputers are the most powerful computing machines on the planet specially designed for immense amount of calculations.
Microcomputer is a relatively small and inexpensive computer.	Supercomputers can occupy few feet to hundreds of feet.
The speed of large computers can be 70 to 100 MIPS.	The processing speed of supercomputers can be 100 to 900 MIPS.
Common applications include word processing, presentations, graphics programs, spreadsheets, and database management systems.	Common applications include mathematical calculations, data analysis, weather forecasting, astronomy, scientific research, data mining, etc. 

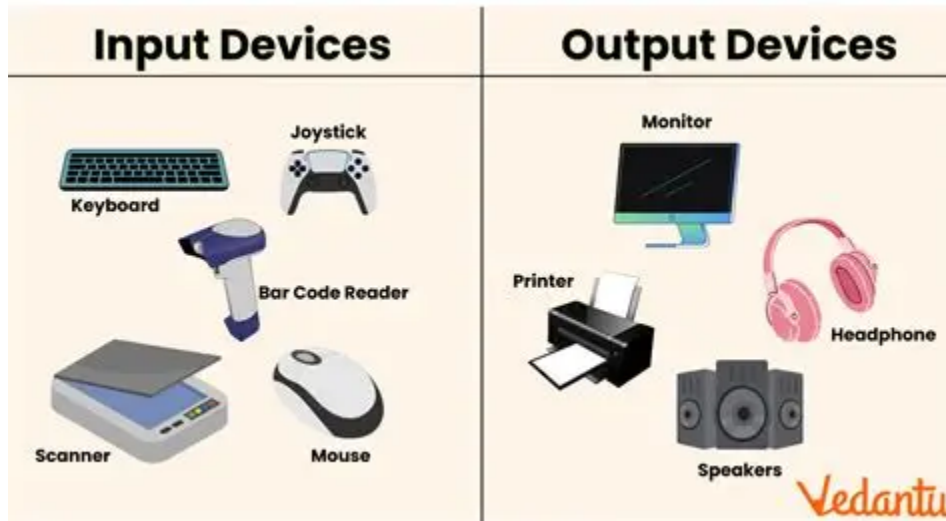
1.4 Components of a Computer System

1. Hardware – CPU, monitor, keyboard.
2. Software – OS + applications.
3. Peopleware – Users.

4. Data – Raw facts.

1.5 Input and Output Devices

- Input: Keyboard, mouse, scanner, webcam.
- Output: Monitor, printer, speakers, projector.

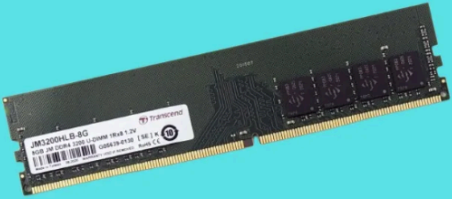
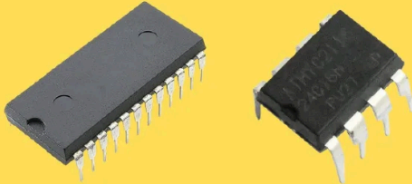


1.6 Memory and Storage

- RAM – Temporary, fast.
- ROM – Permanent, non-volatile.
- Secondary Storage – Hard disk, SSD.
- Portable Storage – USB, CD/DVD.
- Cloud Storage – Online services.

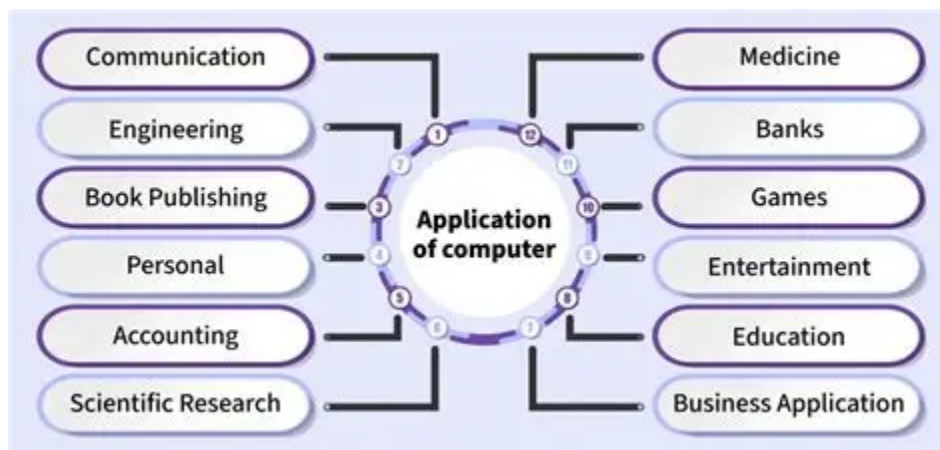
Memory Hierarchy Design



RAM	ROM
	
Temporary storage for active processes	Permanent storage for firmware and critical data
Volatile: Data lost when power is off	Non-volatile: Data retained without power
Ultra-fast: Milliseconds access times	Slower: Access times in microseconds
Easily erased and rewritten	Limited rewriting (e.g., EEPROM, Flash)
Includes DRAM, SRAM, DDR, SDRAM, etc.	Includes PROM, EPROM, EEPROM, Mask ROM
High capacities (GBs), enabling multitasking	Lower capacities (MBs to a few GBs)
Used in Gaming, video editing, software development	Used for BIOS chips, embedded systems, appliances
Generally more cost-effective per GB	More expensive due to complexity
Directly affects system speed and responsiveness	Influences boot times and system stability
Random access to any data	Sequential or random access depending on type

1.7 Applications of Computers

- Education 🎓 – Online classes.
- Business 💼 – Accounting, payroll.
- Healthcare 🏥 – Patient records.
- Government 🏛️ – E-governance.
- Entertainment 🎮 – Movies, gaming.
- Banking 🏦 – Online transactions.



1.8 Advantages and Limitations

Advantages

Speed &
accuracy

Multitasking

Connectivity

Limitations

Cannot think
independently

Dependent on human
input

Risk of misuse
(hacking)

Advantages and Disadvantages of computer

Advantages	Disadvantages
• Creating Tables • Storage Capacity • Accuracy • Multitasking • Learning • Entertainment • Communication • Medical Help • Business	• Increased Screen Time • Expensive to Maintain • Risk of Viruses • Less Social Interaction • Privacy Concerns • Waste of Time • Distraction from Studies • False Information • Hardware Failure